Effect of Supervised Fine-Tuning on Instruction-Tuned LLMs for Hate-Speech Classification

Final Experimental Report

Llama 3.2 3B Instruct and Mistral 7B Instruct v0.3 (4-bit Unsloth variants)

Abstract

This study evaluates supervised fine-tuning (SFT) for three-class hate-speech classification using Llama 3.2 3B Instruct and Mistral 7B Instruct v0.3. Each model was evaluated before and after one epoch of LoRA-based 4-bit fine-tuning on the same HateXplain split. A strict parser was used in the original evaluation. Post-analysis showed that many fine-tuned Mistral responses repeated the same class label, so a deterministic relaxed parser was introduced to separate recoverable formatting artifacts from contradictory outputs. Under the relaxed-parser evaluation, fine-tuned Mistral achieved the strongest aggregate metrics: 60.86% accuracy and 58.87% macro F1, with 11.85% invalid outputs. Fine-tuned Llama achieved 56.91% accuracy and 49.67% macro F1, with only 2.91% invalid outputs. Llama improved strongly for Hate and Normal but degraded for Offensive. The results show that output validity and classification quality should be reported separately in generative classification.

Keywords— hate-speech classification, supervised fine-tuning, LoRA, instruction-tuned language models, deterministic parsing, HateXplain.

1. Introduction

Instruction-tuned language models can perform classification through prompted generation. Their practical performance depends on two abilities: selecting the correct class and producing a response that can be mapped reliably to the required label. This study compares baseline and fine-tuned versions of Llama 3.2 3B Instruct and Mistral 7B Instruct v0.3 on the same three-class HateXplain test set.

The report is based only on the supplied experimental artifacts and the verified post-analysis parser results. External sources document the dataset, models, and software stack [1]–[12]. The main question is whether one epoch of parameter-efficient SFT improves end-to-end classification performance for each model.

2. Experimental Setup

2.1 Dataset and Splits

The run manifests contain 15,383 training examples, 1,922 validation examples, and 1,924 test examples. The test set includes 594 Hate, 782 Normal, and 548 Offensive examples. All four prediction files contain the same test IDs, texts, and true labels, with no duplicate IDs.

2.2 Models and Fine-Tuning

The evaluated repositories were unsloth/Llama-3.2-3B-Instruct-bnb-4bit and unsloth/mistral-7b-instruct-v0.3-bnb-4bit [10], [11]. Fine-tuning used LoRA [3], 4-bit quantization, Unsloth [9], and one SFT epoch. Completion-only loss masked prompt tokens, so only the assistant response contributed to the loss. The base model weights remained frozen; only adapter parameters were trained.

2.3 Evaluation Protocol

Each response was mapped to Hate, Normal, or Offensive. The original evaluation required strict parser compliance. During post-analysis, repeated identical labels were found to account for most fine-tuned Mistral rejections. A deterministic relaxed parser was therefore applied to that condition: if exactly one valid label appears, it is used; repeated instances of the same label are collapsed to one prediction; responses containing two or more different valid labels remain invalid; and responses containing no valid label remain invalid. This post-analysis does not alter contradictory predictions. Accuracy treats invalid outputs as incorrect. Macro precision, recall, and F1 are computed over the three target classes.

3. Overall Results

Model	Stage	Accuracy	Macro P	Macro R	Macro F1	Invalid outputs
Llama 3.2 3B	Baseline	34.41	48.12	34.85	39.19	472 (24.53%)
Llama 3.2 3B	Fine-tuned	56.91	58.98	52.36	49.67	56 (2.91%)
Mistral 7B	Baseline	35.86	51.48	38.96	29.21	41 (2.13%)
Mistral 7B	Fine-tuned	60.86	70.17	57.91	58.87	228 (11.85%)

Table 1. End-to-end test performance using the relaxed-parser result for fine-tuned Mistral (%); N = 1,924 per condition.

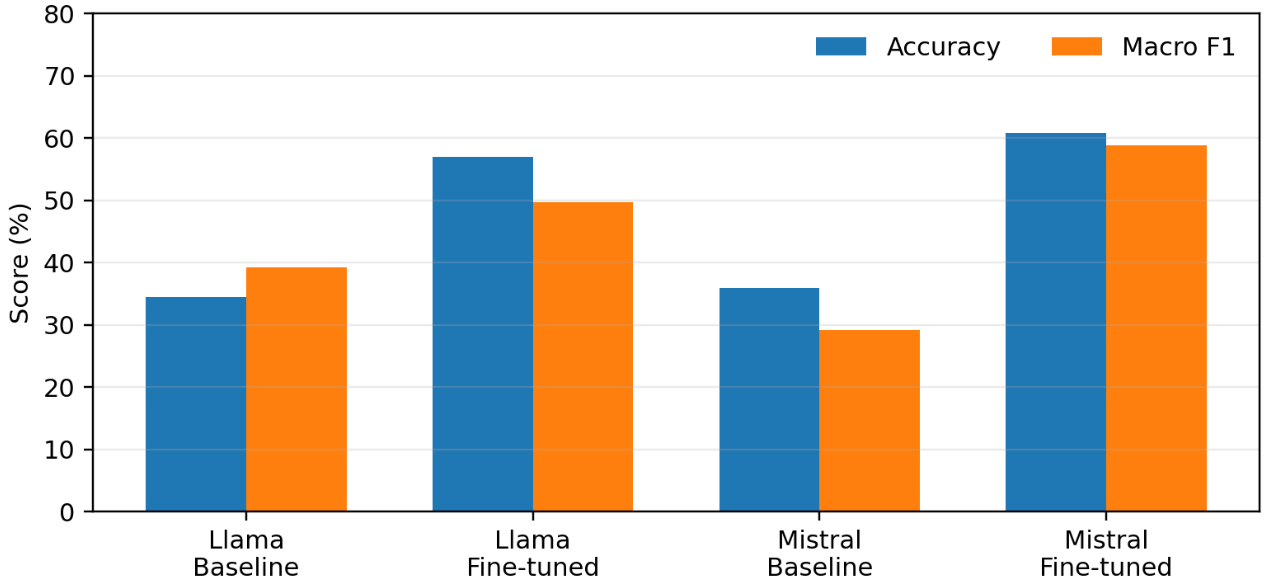

Figure 1. Overall performance using the corrected fine-tuned Mistral evaluation.

3.1 Llama 3.2 3B

Fine-tuning improved every aggregate metric for Llama. Accuracy rose by 22.51 percentage points, from 34.41% to 56.91%. Macro F1 rose by 10.48 points, from 39.19% to 49.67%. Invalid outputs fell from 472 to 56, an 88.14% relative reduction. At item level, 685 baseline errors became correct, while 252 baseline-correct items became incorrect; 410 items were correct in both stages.

3.2 Mistral 7B

The relaxed-parser evaluation changes the interpretation of fine-tuned Mistral. Accuracy increased from 35.86% to 60.86%, and macro F1 increased from 29.21% to 58.87%. Invalid outputs rose from 2.13% to 11.85%, but not to the 90.28% indicated by strict exact matching. Fine-tuned Mistral therefore achieved the strongest aggregate accuracy and macro F1, while fine-tuned Llama retained the lowest invalid-output rate.

4. Class-Level Analysis

Model	Stage	Hate F1	Normal F1	Offensive F1
Llama 3.2 3B	Baseline	37.99	42.60	36.99
Llama 3.2 3B	Fine-tuned	66.29	66.63	16.08
Mistral 7B	Baseline	8.28	32.95	46.41
Mistral 7B	Fine-tuned	73.81	72.09	30.70

Table 2. Per-class F1 scores using the relaxed-parser result for fine-tuned Mistral (%).

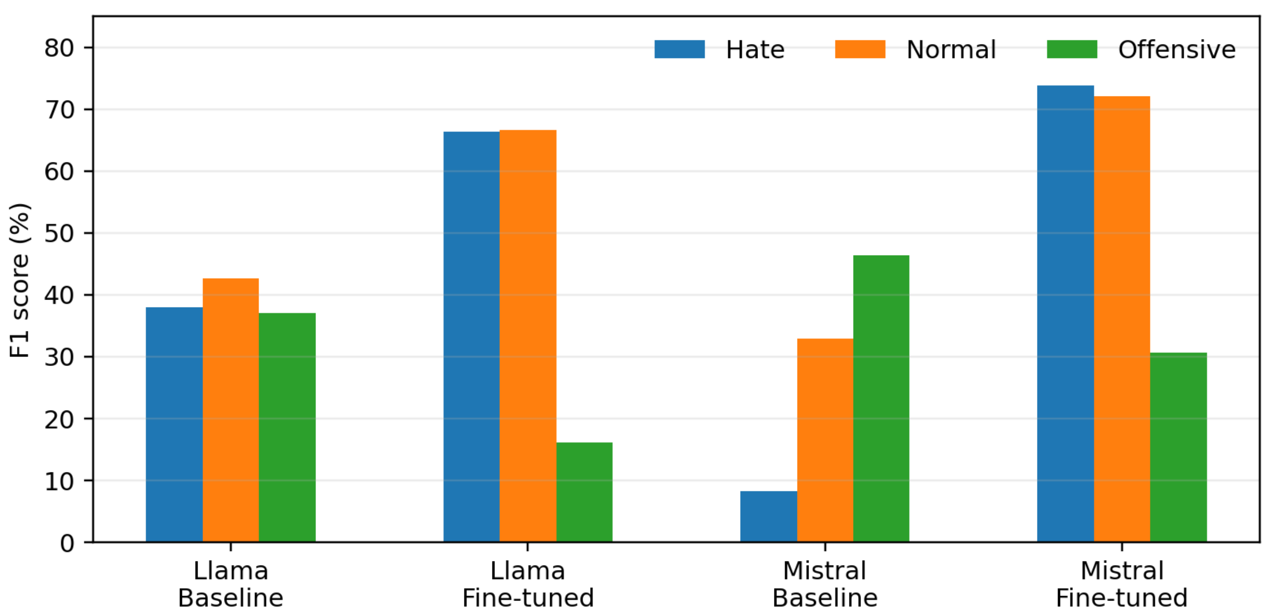

Figure 2. Class-specific F1 scores under the corrected evaluation.

Llama fine-tuning improved Hate recall from 33.67% to 58.92% and Hate F1 from 37.99% to 66.29%. Normal recall increased from 31.46% to 88.49%, and Normal F1 reached 66.63%. The model also shifted strongly toward Normal, predicting that label 1,295 times compared with 373 at baseline.

The main Llama weakness was Offensive. Offensive recall fell from 39.42% to 9.67%, and F1 fell from 36.99% to 16.08% (16.08% in the saved rounded comparison table). Of 548 true Offensive examples, 403 were predicted as Normal and 75 as Hate after fine-tuning. The aggregate improvement therefore came from Hate, Normal, and higher output validity, not balanced improvement across all classes.

Baseline Mistral predicted Offensive 1,624 times. This produced 91.97% Offensive recall but weak Hate and Normal performance. After fine-tuning and relaxed parsing, Hate F1 reached 73.81%, Normal F1 reached 72.09%, and Offensive F1 reached 30.70%. The model improved substantially on Hate and Normal, but Offensive remained its weakest class.

5. Output Validity and Compliance

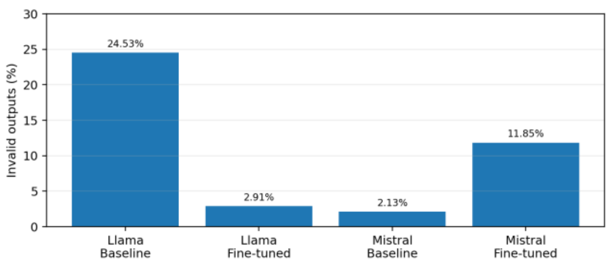

Figure 3. Invalid-output rates after applying the relaxed parser to fine-tuned Mistral.

Output validity is a separate part of end-to-end performance. Fine-tuned Llama reduced invalid outputs from 24.53% to 2.91%. Fine-tuned Mistral had 228 invalid responses under relaxed parsing, equal to 11.85% of the test set. This rate is higher than the Mistral baseline rate of 2.13%, but far lower than the original strict-parser result.

Post-analysis showed that most strict-parser failures were repeated identical labels, such as "Hate" followed by "Hate" or "Normal" followed by "Normal." These outputs express one non-contradictory class decision but violate an exact single-label format. The relaxed parser collapses only repeated instances of the same valid label. Responses containing different labels, partial conflicting labels, or no valid class remain invalid.

The original exact-match result is retained in Appendix A for transparency. It measures strict interface compliance, whereas the relaxed-parser result better represents classification performance when repeated identical labels are treated as recoverable formatting artifacts.

6. Artifact Verification and Reproducibility

All prediction files contain 1,924 rows with aligned test IDs and true labels. Saved baseline and Llama metrics were reproduced from the prediction files. The relaxed Mistral metrics were independently recomputed from the raw outputs using the stated deterministic rules.

The archives include predictions, metrics, error subsets, invalid-output subsets, comparison tables, confusion matrices, and run manifests. The parser rules and the preserved exact-match appendix make the revised evaluation auditable.

7. Discussion

Fine-tuning improved the aggregate classification metrics of both models under the reported parsing procedures, but in different ways. Fine-tuned Mistral achieved the highest aggregate accuracy (60.86%) and macro F1 (58.87%). Fine-tuned Llama achieved slightly lower aggregate scores but the best output validity, with only 2.91% invalid responses. These findings replace the earlier conclusion that fine-tuned Llama was the strongest overall condition.

For Mistral, the main observed issue was an end-to-end output-format compliance failure under strict exact-match evaluation, not necessarily a complete semantic failure. Repeated identical labels were recoverable because they represented one class decision. Conflicting labels were not recoverable and remained invalid. The relaxed-parser evaluation therefore separates formatting artifacts from genuine ambiguity.

Generative classification should be evaluated on two dimensions. Classification metrics describe the quality of mapped label decisions, while invalid-output rates describe compliance with the required response schema. Reporting both prevents a parser choice from being mistaken for model-level semantic performance.

8. Limitations

- Only one epoch and one fine-tuning configuration were evaluated; no seed or hyperparameter sensitivity analysis was available.
- The study used one fixed train/validation/test split, so variation across alternative splits is unknown.
- Parser design affects end-to-end scores. The relaxed parser is deterministic and conservative, but it was introduced after inspection of the fine-tuned Mistral outputs and should be treated as a post-analysis correction.
- The archives did not contain training logs, adapters, checkpoints, or hardware timing data.
- The outputs support diagnosis of the Mistral formatting artifact, but they do not establish why fine-tuning caused repeated labels.

9. Conclusion

Fine-tuned Mistral achieved the strongest aggregate performance after deterministic relaxed parsing, with 60.86% accuracy, 70.17% macro precision, 57.91% macro recall, and 58.87% macro F1. Its invalid-output rate was 11.85%. Fine-tuned Llama reached 56.91% accuracy and 49.67% macro F1, while producing the lowest invalid rate at 2.91%. Llama improved strongly for Hate and Normal but degraded on Offensive. The revised findings show that strict output compliance and classification quality are related but distinct: repeated identical labels can cause severe exact-match penalties without representing contradictory predictions.

References

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Appendix A. Original Exact-Match Evaluation

The original strict parser required one exact class label. Under that evaluation, fine-tuned Mistral produced 1,737 invalid outputs (90.28%), with 8.11% accuracy, 27.81% macro precision, 6.65% macro recall, and 10.73% macro F1. These results are preserved for transparency and reproducibility. They primarily reflect repeated identical labels and related formatting artifacts. The relaxed-parser evaluation in the main report is used for substantive model comparison.

Model	Stage	Accuracy	Macro P	Macro R	Macro F1	Invalid outputs
Mistral 7B	Fine-tuned	8.11	27.81	6.65	10.73	1,737 (90.28%)

Table A1. Original exact-match fine-tuned Mistral results (%).